

Basic Quantum Algorithms

Thomas Konrad @QCSpringSchool Alpine Heath
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Content

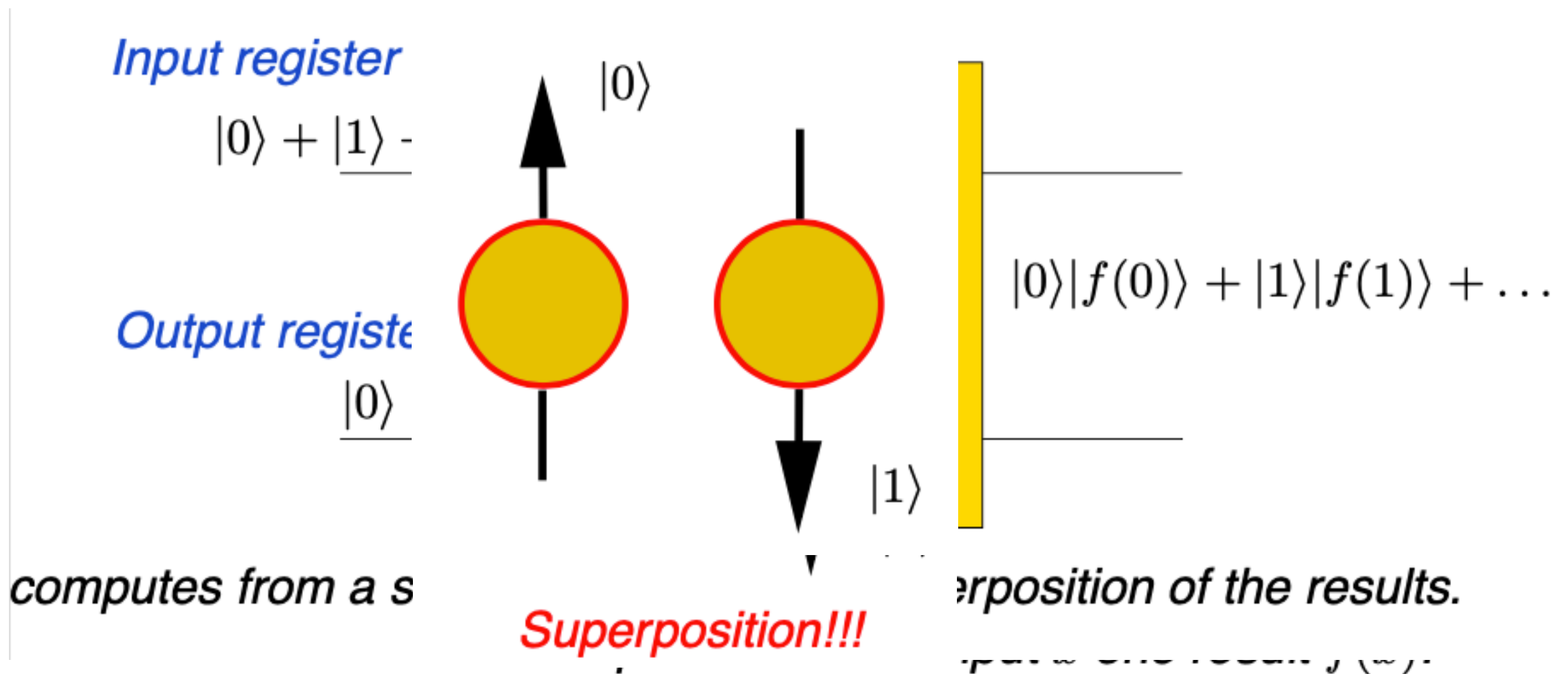
- Deutsch Algorithm:
First Quantum Algorithm!
- Deutsch Josza Algorithm:
First exponential speed up!
- Grover Algorithm:
The workhorse of QC!



CARTIN—CASS 45	
CASEY Robt L 32 Sweet Cr Ded.	326-2370
Robt & Lauren B 115 Av Ded.	326-0635
Robt W 71 Herbert Rd Bra	889-5137
Rose 617 Broad Way	331-6948
Ruth B 208 Atchc Av Hst	925-2525
Sean 141 Carrol Av Wswd	329-9412
Stacie 30 Chapman Wey	331-3652
Stephen & Katherine	
570 Middle Wey	
Stephen R 17 Draper Cn	340-6658
Theresa 17 Bismarck Ln Qui	828-1108
Thos A Jr 194 School Wswd	479-4923
Thos R 1 Wentworth Rd Can	328-6078
Timothy P 25 French Cui	828-2474
Tom & Judy 637 Pleasant Mt	326-6969
V 9 Fore River Av Wey	326-1140
Valter J Jr 17 Aquson Cr Hin	328-4662
m C 212 Central Av Humrck	328-0734
11 36 Bayley Wswd	331-5746
ten & Bill 11 Knn	
968 Auant	

Working Principle

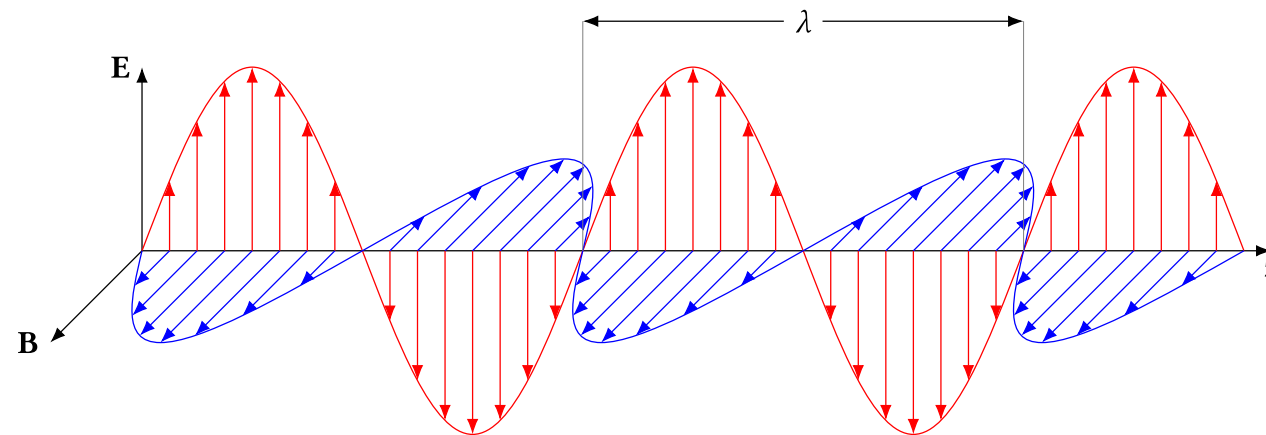
A **qubit**, can be stored in a quantum mechanical system, for example a system carrying a spin:



Conventional Computer	Quantum Computer
A bit represents either 0 or 1 (uncharged or charged capacitor). Three bits: Either 000 or 001 or 010... or 111 at a given time. Register with n bits represents one of the 2^n numbers at time t.	A qubit with $\psi\rangle = \alpha_0 0\rangle + \alpha_1 1\rangle$, $\alpha_i \neq 0$, represents both 0 and 1. Three qubits in state $\Psi\rangle = \alpha_0 000\rangle + \alpha_1 001\rangle + \alpha_2 010\rangle \dots + \alpha_7 111\rangle$, $\alpha_i \neq 0$ represent eight numbers simultaneously. n qubits with $\Psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i i\rangle$ represent 2^n numbers at time t.

What is quantum information?

Light is a EM wave

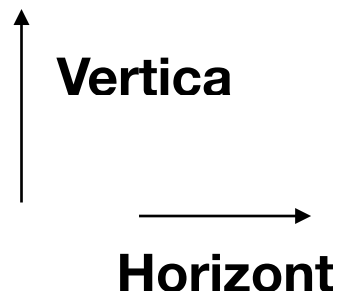


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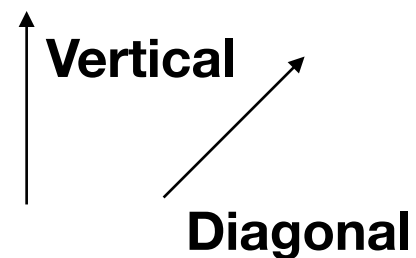
Electric field separates charges, i.e.

polarises

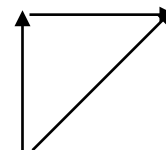
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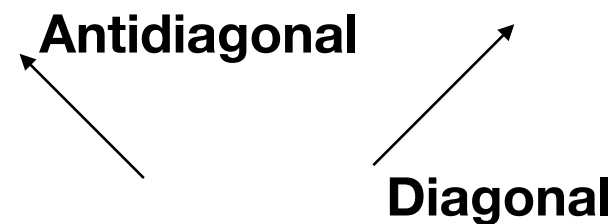
Distinguish by means of filters



Not distinguishable for single photon



Diagonal is a superposition of Vertical and Horizontal



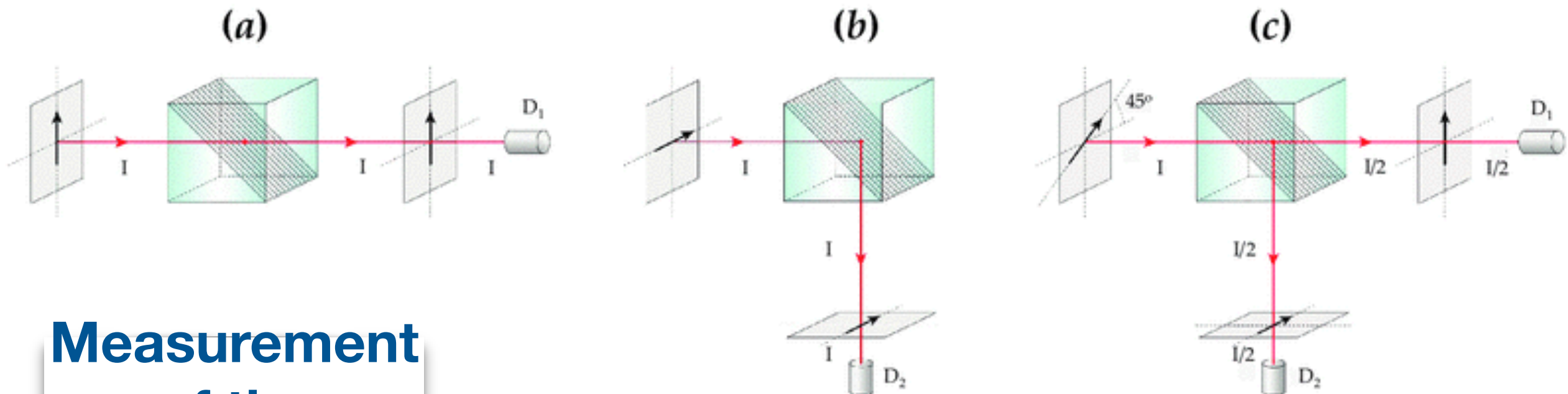
Diagonal and Antidiagonal have no common component

How do we understand quantum information

Can send V (“0”) or H (“1”), or “superposition” of V (“0”) and H (“1”).

Quantum Bit $0 + 1$

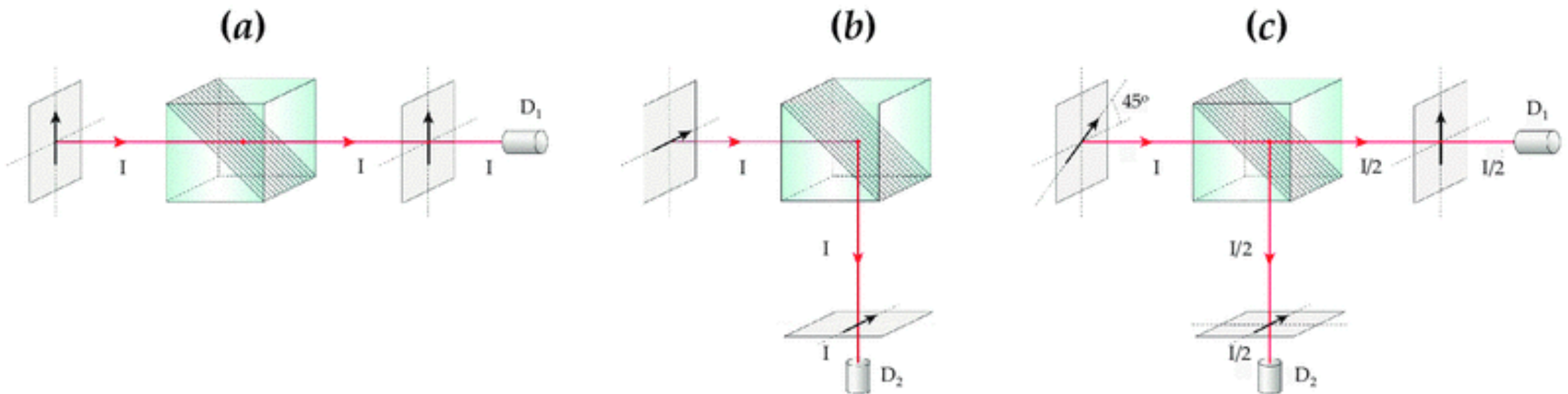
A photon is an elementary excitation of a light wave



**Measurement
of the
polarisation**

How do we understand quantum information
Can send V (“0”) or H (“1”), or “superposition” of V (“0”) and H (“1”).

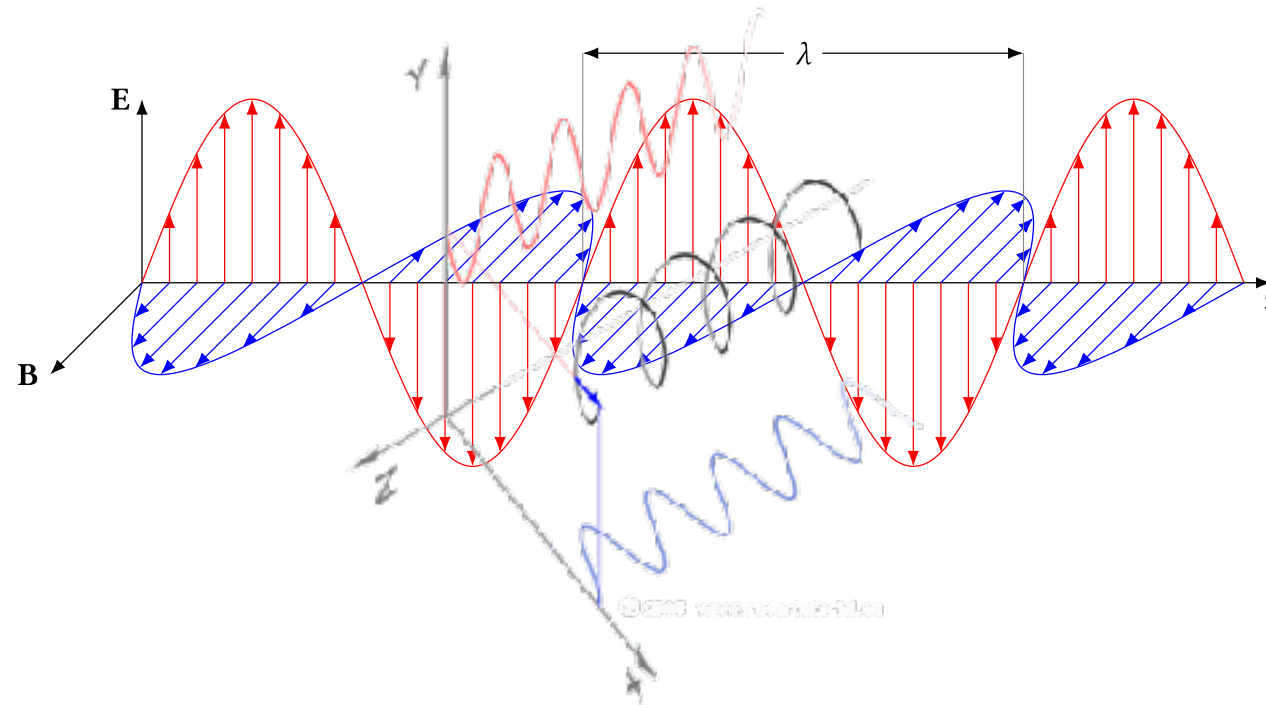
Quantum Bit $|0\rangle + |1\rangle$



Can detect only $|0\rangle$ or $|1\rangle$, not $|0\rangle + |1\rangle$

How do we understand quantum information

A photon is an elementary excitation of a light wave

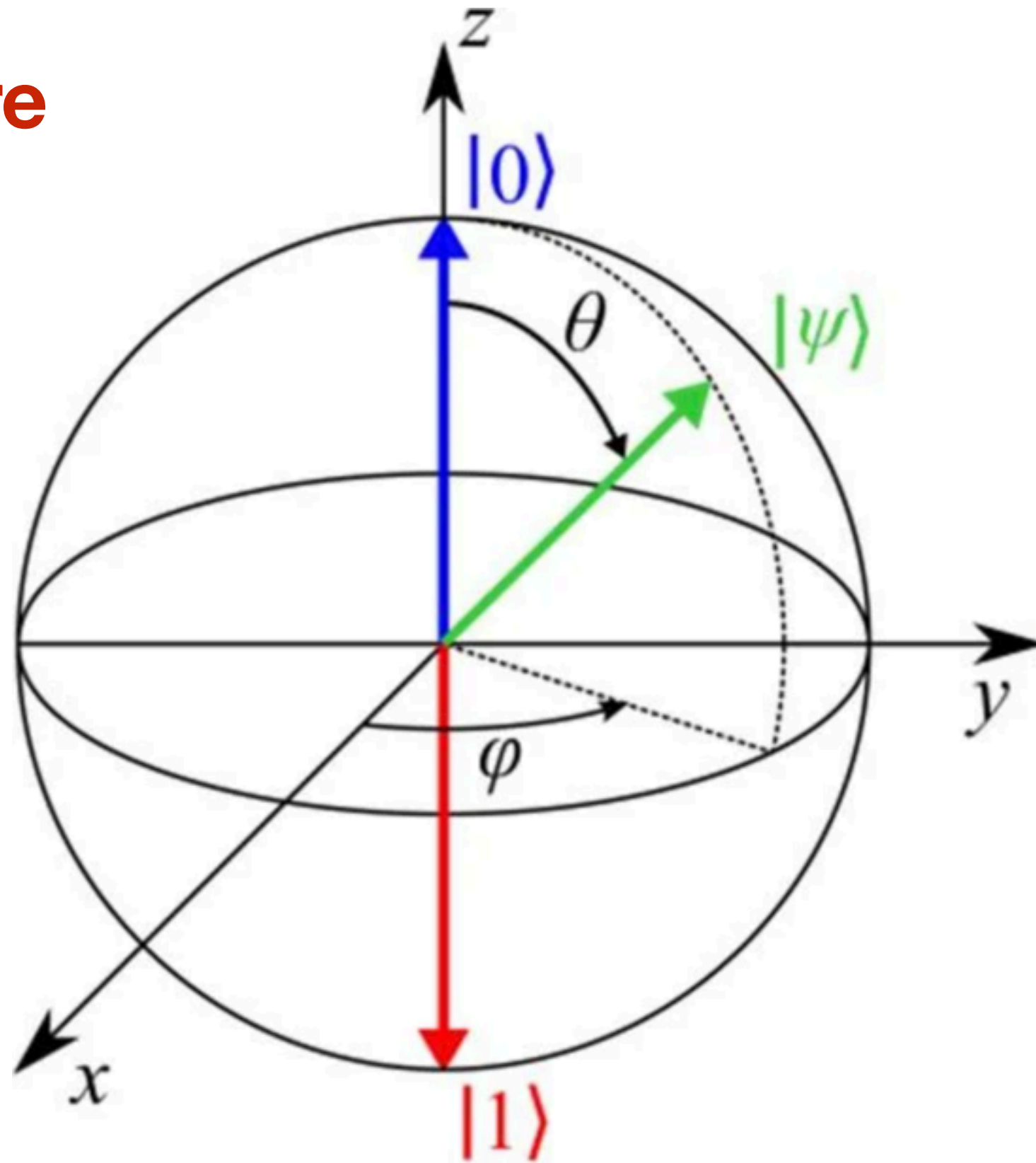


circular: $\mathbf{E}_r(\mathbf{x}, t) = g(\mathbf{x}) \left(\exp(i\omega t) \mathbf{h} + \exp\left\{i\left(\omega t + \frac{\pi}{2}\right)\right\} \mathbf{v} \right)$
 $= g(\mathbf{x}) \exp(i\omega t) (\mathbf{h} + i\mathbf{v})$

$$|r\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

Qubit: $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$

Bloch Sphere



$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

Qubit: $|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle \in \mathcal{H} \iff \langle\psi| = c_0^* \langle 0| + c_1^* \langle 1| \in \hat{\mathcal{H}}$
 $|\varphi\rangle = d_0 |0\rangle + d_1 |1\rangle$

Inner Product : $\langle\psi| (|\phi\rangle) = \langle\psi|\phi\rangle \in \mathbb{C}$ **Computational basis :** $\langle i|j\rangle = \delta_{ij}$

$$= (c_0^* \langle 0| + c_1^* \langle 1|)(d_0 |0\rangle + d_1 |1\rangle) = c_0^* d_0 \langle 0|0\rangle + c_0^* d_1 \langle 0|1\rangle + \dots = c_0^* d_0 + c_1^* d_1$$

Outer Product : $|\varphi\rangle\langle\psi| (|\phi\rangle) \Rightarrow |\varphi\rangle \langle\psi|\phi\rangle = \alpha |\varphi\rangle$ with $\alpha = \langle\psi|\phi\rangle$

For $|v\rangle, |w\rangle \in \mathcal{H}$:

Projector: $|v\rangle \rightarrow |w\rangle \langle w| (|v\rangle) \equiv |w\rangle \langle w|v\rangle = \alpha |w\rangle$ with $\alpha = \langle w|v\rangle$

Basis representation: $|u\rangle = \sum_{i=0}^{\infty} \alpha_i |i\rangle = \sum_{i=0}^{\infty} |i\rangle (\langle i|u\rangle) = \left(\sum_{i=0}^{\infty} |i\rangle \langle i| \right) |u\rangle$ for all $|u\rangle \in \mathcal{H}$
 $\Rightarrow \sum_{i=0}^{\infty} |i\rangle \langle i| = \mathbb{I}$

$$A = \mathbb{I} A \mathbb{I} = \left(\sum_{i=1}^{\infty} |i\rangle \langle i| \right) A \left(\sum_{j=1}^{\infty} |j\rangle \langle j| \right) = \sum a_{ij} |i\rangle \langle j|$$

Spectral decomposition: $A = \sum_a a |a\rangle \langle a|$ $f(A) := \sum_a f(a) |a\rangle \langle a|$

Application of Outer Product :

Bit Flip: $|0\rangle \xrightarrow{X} |1\rangle, X|0\rangle = |1\rangle, |1\rangle\langle 0|(|0\rangle) = |1\rangle\langle 0|0\rangle = |1\rangle$

$$|1\rangle \xrightarrow{X} |0\rangle, X|1\rangle = |0\rangle, |0\rangle\langle 1|1\rangle = |0\rangle$$

Write X with outer products of the computational basis states!

$X = ?$ Answer: $X = |1\rangle\langle 0| + |0\rangle\langle 1|$

Bit-Phase Flip: $|0\rangle \xrightarrow{Y} -i|1\rangle \quad Y = ?$

$$|1\rangle \xrightarrow{Y} i|0\rangle \quad Y = -i|1\rangle\langle 0| + i|0\rangle\langle 1|$$

Phase Flip: $|0\rangle \xrightarrow{Z} |0\rangle, Z|0\rangle = |0\rangle,$

$$|1\rangle \xrightarrow{Z} (-|1\rangle), Z|1\rangle = -|1\rangle,$$

$Z = ?$ Answer: $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$

The identity operator $\mathbb{I}$ and $X \equiv \sigma_1$, $Y \equiv \sigma_2$, $Z \equiv \sigma_3$, form a basis on $\mathcal{L}(\mathcal{H}_2)$

$$A = \frac{1}{\sqrt{2}} \sum_{\mu=0}^3 \alpha_{\mu} \sigma_{\mu} \text{ with } \sigma_0 := \mathbb{I} \text{ and } \alpha_{\mu} = \frac{1}{\sqrt{2}} \langle \sigma_{\mu}, A \rangle = \frac{1}{\sqrt{2}} \text{Tr}[\sigma_{\mu} A]$$

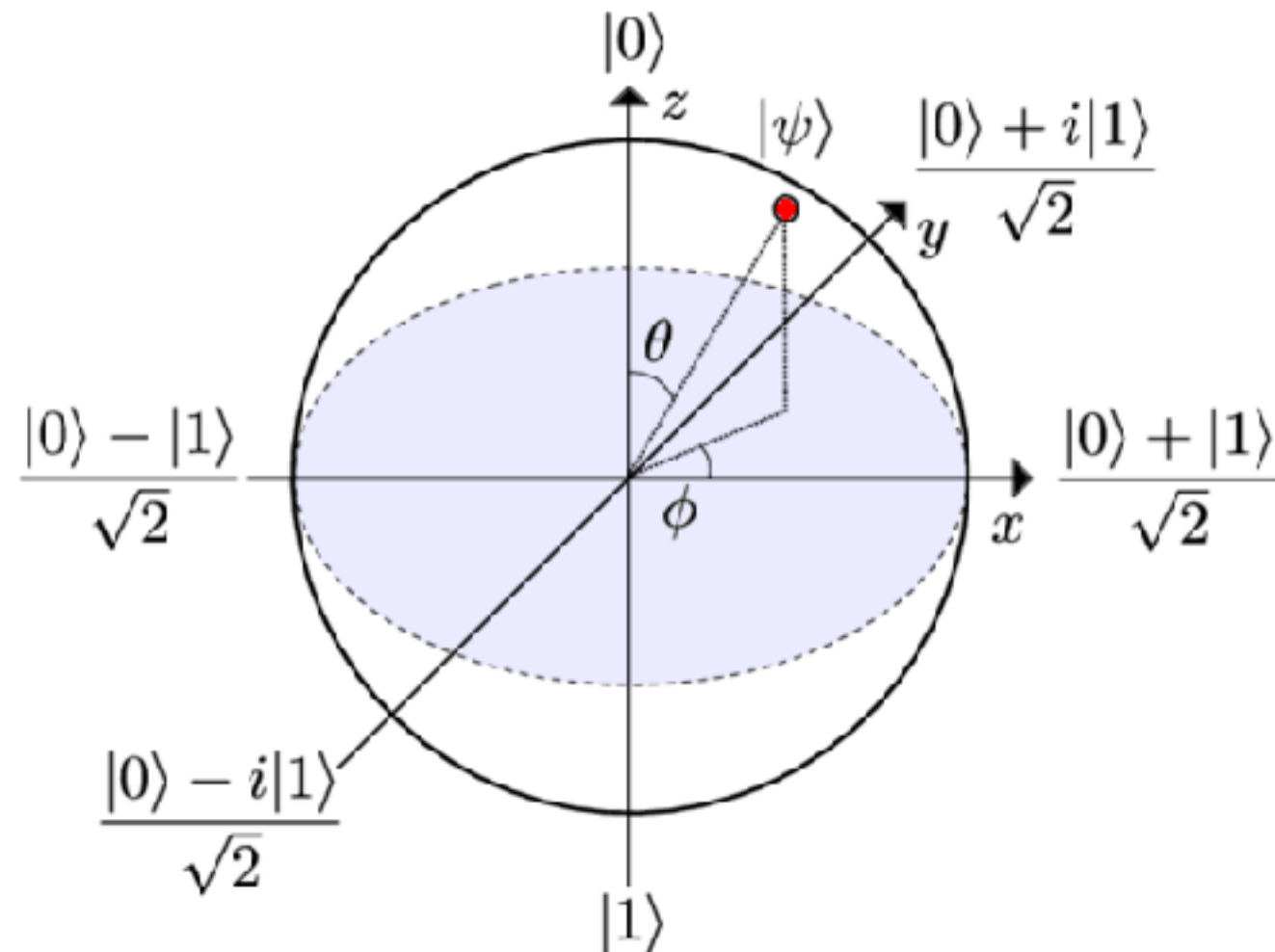
$$|\psi\rangle\langle\psi| = \frac{1}{2}\mathbb{I} + \frac{1}{2} \sum_{k=1}^3 \text{tr}[|\psi\rangle\langle\psi|\sigma_k] \sigma_k = \frac{1}{2}(\mathbb{I} + \vec{r}\vec{\sigma}) \text{ with } \vec{r} = \text{tr}[|\psi\rangle\langle\psi|\vec{\sigma}] \in \mathbb{R}^3.$$

$$\vec{r} = \text{Tr}[|\psi\rangle\langle\psi|\vec{\sigma}] = \langle\psi|\vec{\sigma}|\psi\rangle \hat{=} \begin{pmatrix} \langle\psi|\sigma_1|\psi\rangle \\ \langle\psi|\sigma_2|\psi\rangle \\ \langle\psi|\sigma_3|\psi\rangle \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}.$$

$\vec{r} \cdot \vec{\sigma} |\psi_{\pm\vec{r}}\rangle = \pm |\psi_{\pm\vec{r}}\rangle$ where $|\psi_{\pm\vec{r}}\rangle$ is the state with the Bloch vector

$$\vec{r} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$$

i.e. $|\psi_{\vec{r}}\rangle = |\psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$.



For any unitary gate $U : \mathcal{H}_2 \rightarrow \mathcal{H}_2$, there exists a unique $\vec{r} \in \mathbb{R}^3$ with $||\vec{r}|| = 1$ and $\alpha, \phi \in [0, 2\pi]$, such that

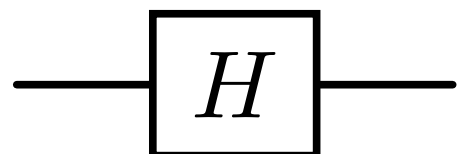
$$U = e^{i\alpha} e^{i\phi \vec{r} \vec{\sigma}} = e^{i\alpha} (\cos \phi \mathbb{I} + i \sin \phi \vec{r} \vec{\sigma}).$$

Any qubit gate $U : \mathcal{H}_2 \rightarrow \mathcal{H}_2$ can be realised up to a global phase factor by three rotations about two axes:

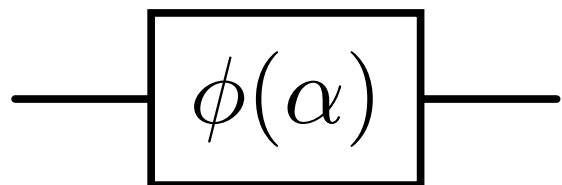
$$\exists \kappa, \lambda, \nu, \mu \in [0, 2\pi[\text{ such that } U = e^{i\kappa} e^{-\frac{i}{2}\lambda\sigma_z} e^{-\frac{i}{2}\mu\sigma_y} e^{-\frac{i}{2}\nu\sigma_z}.$$

Universal Set of Gates:

Any one qubit gate can be generated by a combination of two gates

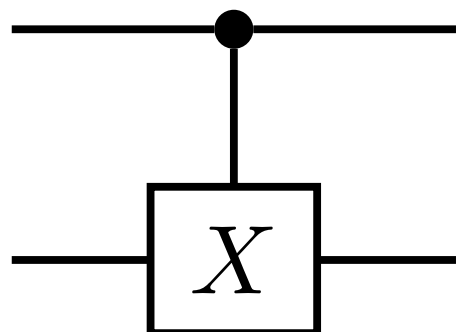


Hadamard gate



$$\begin{aligned} |0\rangle &\rightarrow e^{i\alpha} |0\rangle, \\ |1\rangle &\rightarrow |1\rangle \end{aligned} \quad \text{Phase gate}$$

Any gate transforming $n \in \mathbb{N}$ qubits can be generated by a combination with



Controlled-Bitflip gate

Deutsch Algorithm

First Evidence for Power of Quantum Computing: Number of Calls (queries) to characterise Boolean functions.

Simple Example: $f : \{0, 1\} \rightarrow \{0, 1\}$

Is $f(0) = f(1)$ or $f(0) \neq f(1)$?

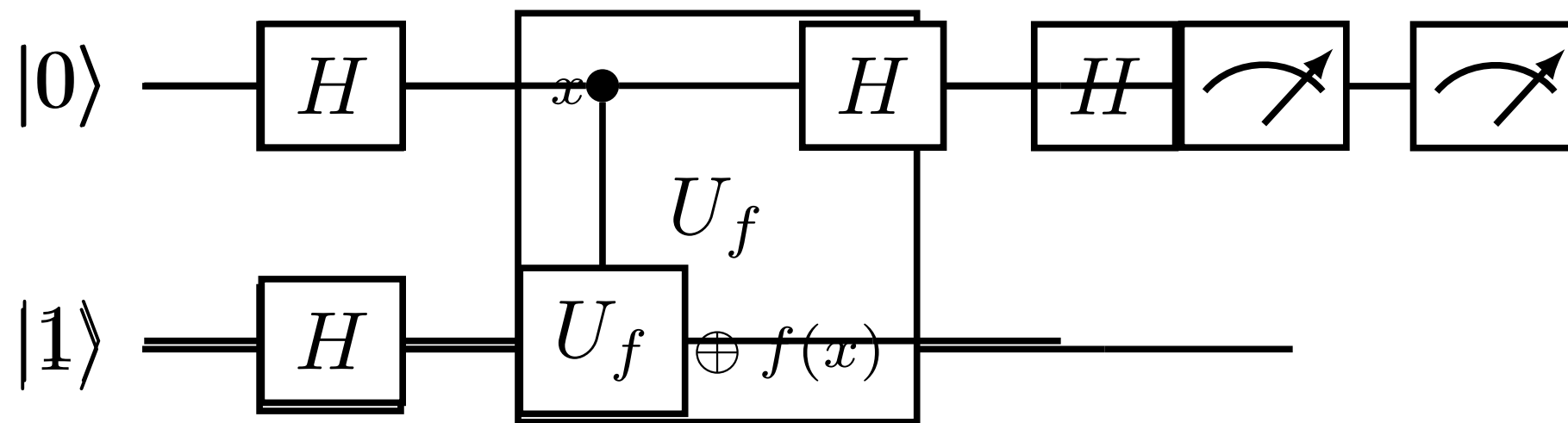
**Imagine you have to pay for each evaluation of the function 20 Rand!
How much do you have to pay?**

Classically both functional values are needed to decide:



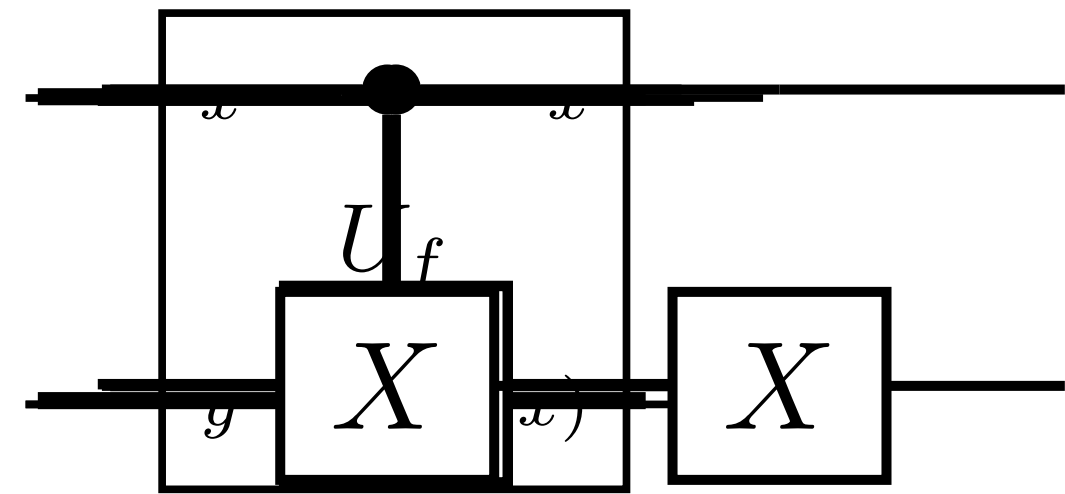
Deutsch Algorithm

David Deutsch (1985). Improved: Cleve et al. Proc R. Soc A 454 339 (1998)



$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}} (|0\rangle\langle 0| + |1\rangle\langle 1|) |0\rangle \langle 1| + |1\rangle\langle 0| |1\rangle \langle 1|$$

Computing $f(x)$ on a quantum computer



$$|x\rangle |y\rangle \xrightarrow{U_f} |x\rangle |y \oplus f(x)\rangle$$



$$f(x) = 0 : \quad |x\rangle |y\rangle \xrightarrow{U_f} |x\rangle |y \oplus 0\rangle = |x\rangle |y\rangle$$

$$f(x) = 1 : \quad |x\rangle |y\rangle \xrightarrow{U_f} |x\rangle |y \oplus 1\rangle = |x\rangle X |y\rangle$$

$$f(x) = x : \quad |x\rangle |y\rangle \xrightarrow{U_f} |x\rangle |y \oplus x\rangle = |x\rangle X^x |y\rangle$$

$$\begin{aligned} f(x) = x \oplus 1 : \quad & |x\rangle |y\rangle \xrightarrow{U_f} |x\rangle |y \oplus (x \oplus 1)\rangle \\ & = |x\rangle |(y \oplus x) \oplus 1\rangle = |x\rangle X X^x |y\rangle = |x\rangle X^{x+1} |y\rangle \end{aligned}$$

for any f : $|x\rangle |y\rangle \xrightarrow{U_f} |x\rangle |y \oplus f(x)\rangle = |x\rangle X^{f(x)} |y\rangle$

$$|x\rangle |-\rangle \xrightarrow{U_f} |x\rangle X^{f(x)} |-\rangle = |x\rangle (-1)^{f(x)} |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

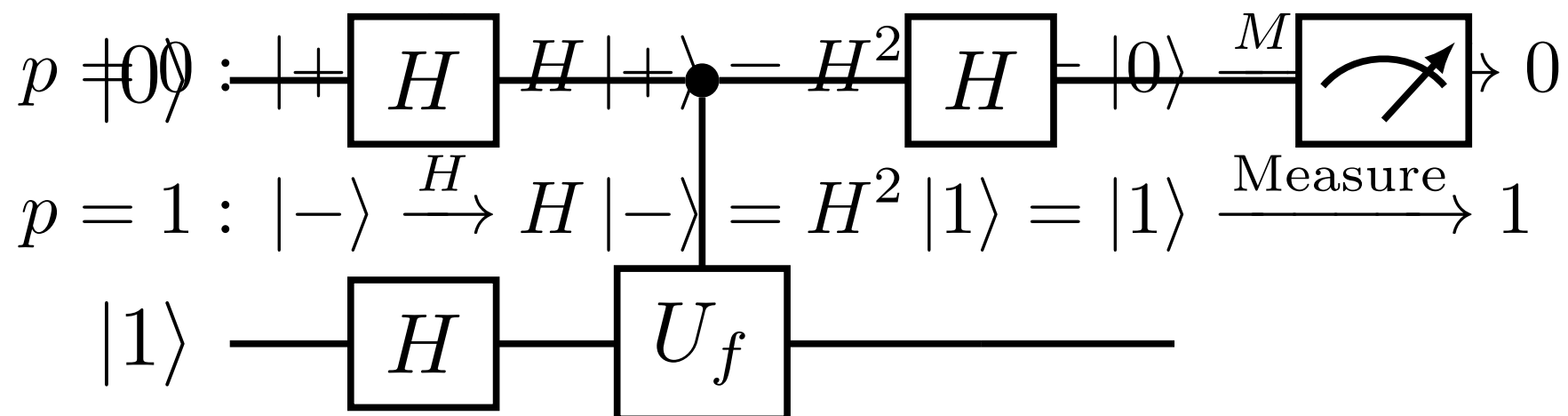
$$\begin{aligned} \Rightarrow \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |y\rangle &\xrightarrow{U_f} \frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle \right) |y\rangle \\ &= \frac{(-1)^{f(0)}}{\sqrt{2}} \left(|0\rangle + (-1)^{(f(1)-f(0))} |1\rangle \right) |y\rangle \end{aligned}$$

for $f(0) = f(1)$: $|+\rangle |-\rangle \xrightarrow{U_f} |+\rangle |-\rangle$

and for $f(0) \neq f(1)$: $|+\rangle |-\rangle \xrightarrow{U_f} |-\rangle |-\rangle$

One could omit the second qubit. Use phase flip Z : $|+\rangle \rightarrow Z^{(f(1)-f(0))} |+\rangle$

Hadamard gate + Measurement results in value of parity $p \equiv f(1) - f(0)$.



Deutsch-Josza Algorithm

Extension to Boolean functions

$$f : \{0, 1 \dots N - 1\} \rightarrow \{0, 1\} \text{ with } N = 2^n$$

Promise: the function is constant or balanced

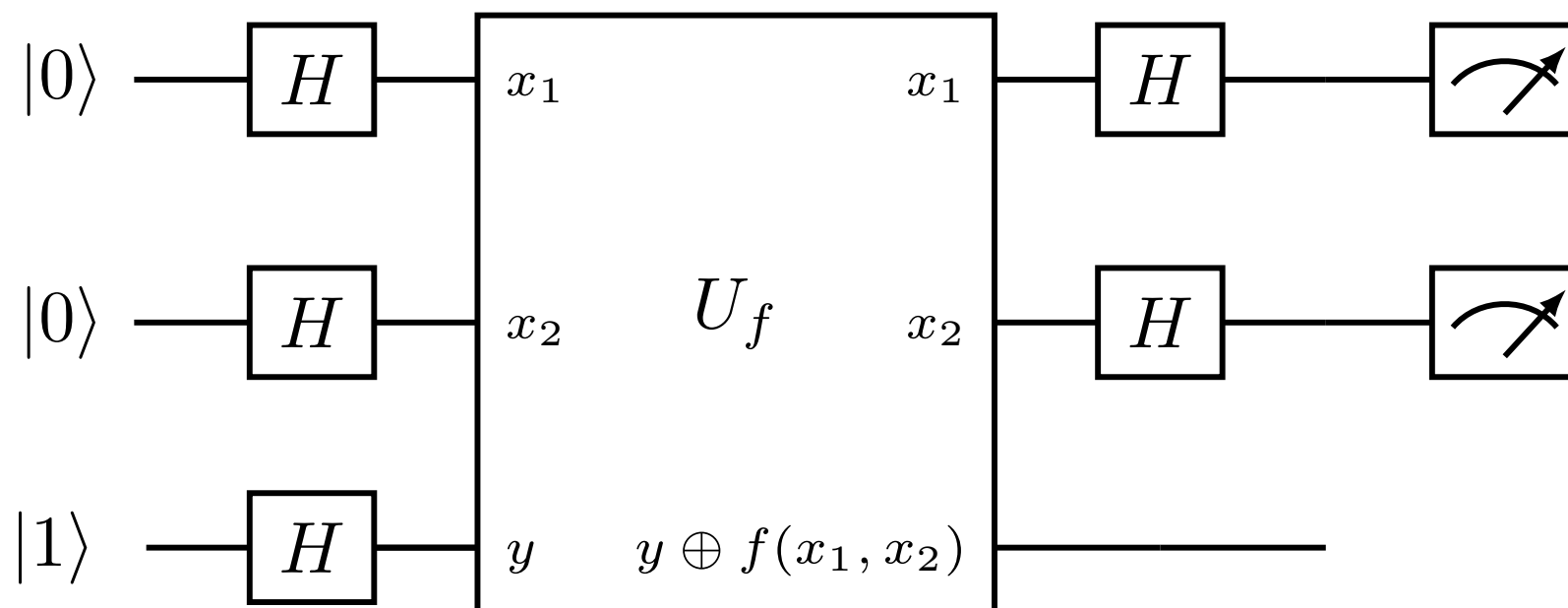
f is constant: for all x -values: $f(x) = 0$ or $f(x) = 1$,

f is balanced: for $N/2$ x -values $f(x) = 0$ and for $N/2$ x -values $f(x) = 1$.

How many queries are needed (each R 20) to determine which is the case?

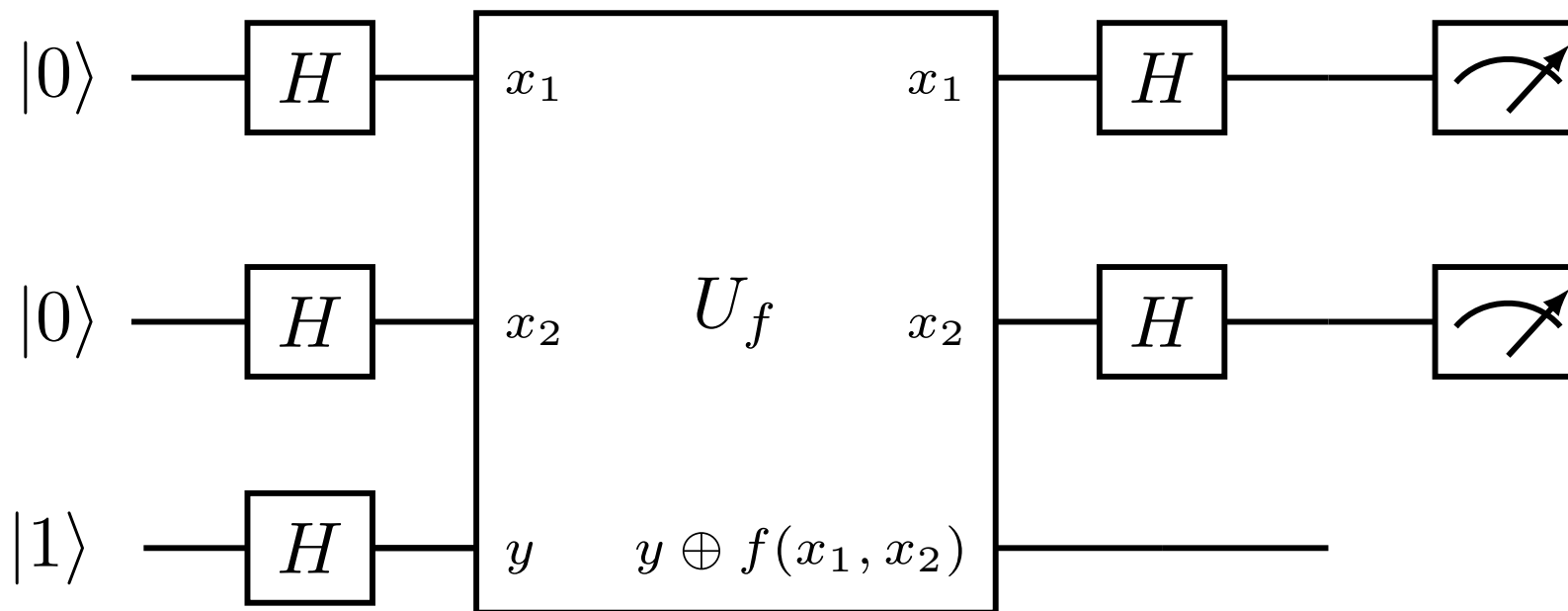
Classically in the worst case: $N/2 + 1$ queries!

For $n = 200$ bit, this is $2^{200} \approx 10^{60}$ steps! Classically intractable!



Quantum: 1 query!

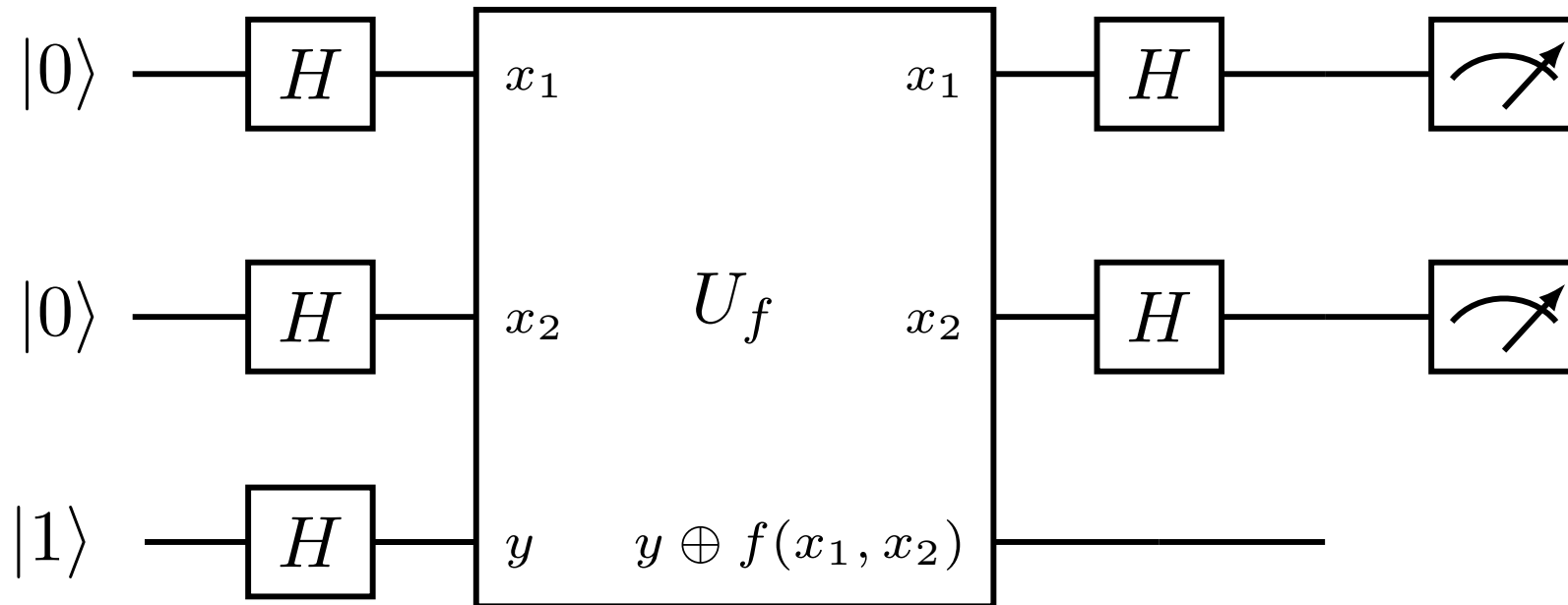
Deutsch-Josza Algorithm



$$|0\rangle |0\rangle \dots |0\rangle |1\rangle \xrightarrow{H^{\otimes n} H^{\otimes n+1}} \frac{1}{\sqrt{2^n}} (|0\rangle + |1\rangle) \dots (|0\rangle + |1\rangle)$$

$$\xrightarrow{U_f} \frac{1}{\sqrt{2^n}} \left(\sum_{x=0}^{2^n-1} |x\rangle \otimes |f(x)\rangle \right) = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \otimes |-\rangle \equiv \sum_{x=0}^{2^n-1} |x\rangle$$

$$\xrightarrow{H^{\otimes n}} = \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \left(\sum_{z=0}^{2^n-1} (-1)^{x \cdot z} |z\rangle \right) |-\rangle$$



$$\frac{1}{\sqrt{2}} \sum_{x_0, \dots, x_{N-1}} |x_0, x_1 \dots x_{N-1}\rangle \xrightarrow{H^{\otimes n}} \frac{1}{2^n} \sum_{x_0, \dots, x_{N-1}} \sum_{z_0, \dots, z_{N-1}} (-1)^{(z_0 x_0 + \dots + z_{N-1} x_{N-1})} |z_0, \dots, z_{N-1}\rangle$$

$$\equiv \frac{1}{2^n} \sum_x \sum_z (-1)^{x \cdot z} |z\rangle$$

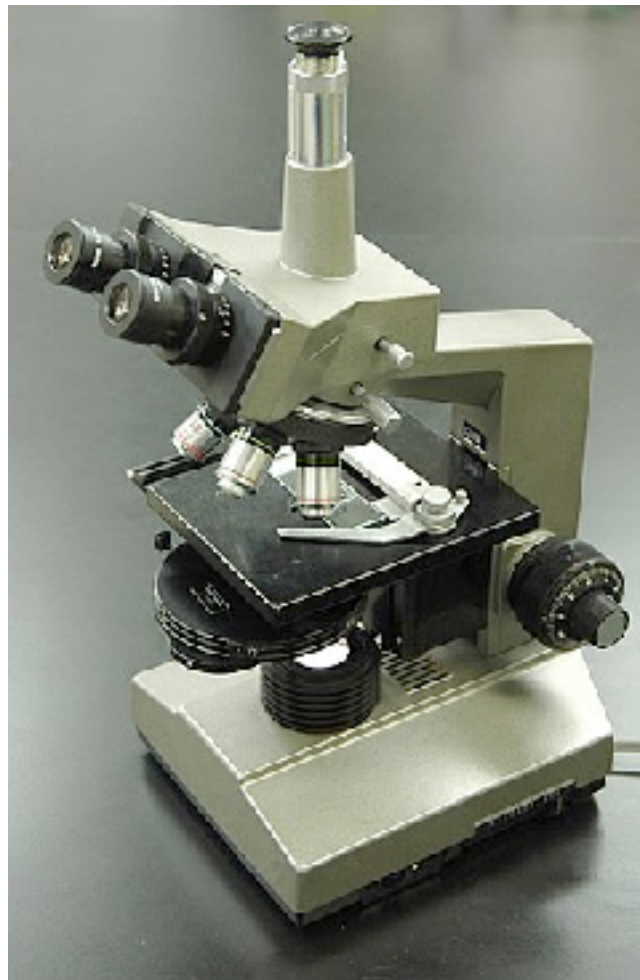
Again last step:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \xrightarrow{H^{\otimes n}} \frac{1}{2^n} \sum_{x=0}^{2^n-1} \sum_{z=0}^{2^n-1} (-1)^{x \cdot z + f(x)} |z\rangle ,$$

If $f(x) = \text{const}$: x-register is in state $|0, \dots, 0\rangle$, i.e. measurement yields $0 \dots 0$

If $f(x)$ balanced : no component of $|0, \dots, 0\rangle$, i.e. measurement does not yield $0 \dots 0$

Grovers Search Algorithm



Optical Implementation of Grover Search

- Find solution to computational problem.

E.g. $1 < p, q < N$ with $pq = N$

- Produce superposition of states that represent possible solutions

$$|\Psi\rangle = \frac{1}{N-1} \sum_{x,y=1}^{N-1} |x\rangle |y\rangle \equiv \frac{1}{N-1} \sum_{x,y=1}^{N-1} |x, y\rangle$$

- Calculate witness $f(x,y) = xy \bmod N$

$$\begin{aligned} |\Psi\rangle &= \frac{1}{N-1} \sum_{x,y=1}^{N-1} |x, y\rangle |0\rangle \\ &\rightarrow \frac{1}{N-1} \sum_{x,y=1}^{N-1} |x, y\rangle |x \cdot y \bmod N\rangle \end{aligned}$$

- NP problem: There exists a polynomial witness f : If x is solution, then $f(x)=0$ and this can be evaluated in polynomial time, i.e with $O(n^k)$ for some k . Assume that we can efficiently compute

$$\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle |0\rangle \xrightarrow{U_f} \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle |f(x)\rangle$$

- Measurement in second register

$$\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle |f(x)\rangle \xrightarrow{\mathbb{I} \otimes |0\rangle\langle 0|} \frac{1}{\sqrt{N}} |x_s\rangle |f(x_s) = 0\rangle$$

- Success probability $p = 1/N$ for N candidates. This is the same as for random guess.
- Aim: amplify amplitude of solution

$$|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle |f(x)\rangle$$

$$\xrightarrow{M} \alpha |x_s\rangle |f(x_s = 0)\rangle + \sum \beta_x |x\rangle |f(x)\rangle$$

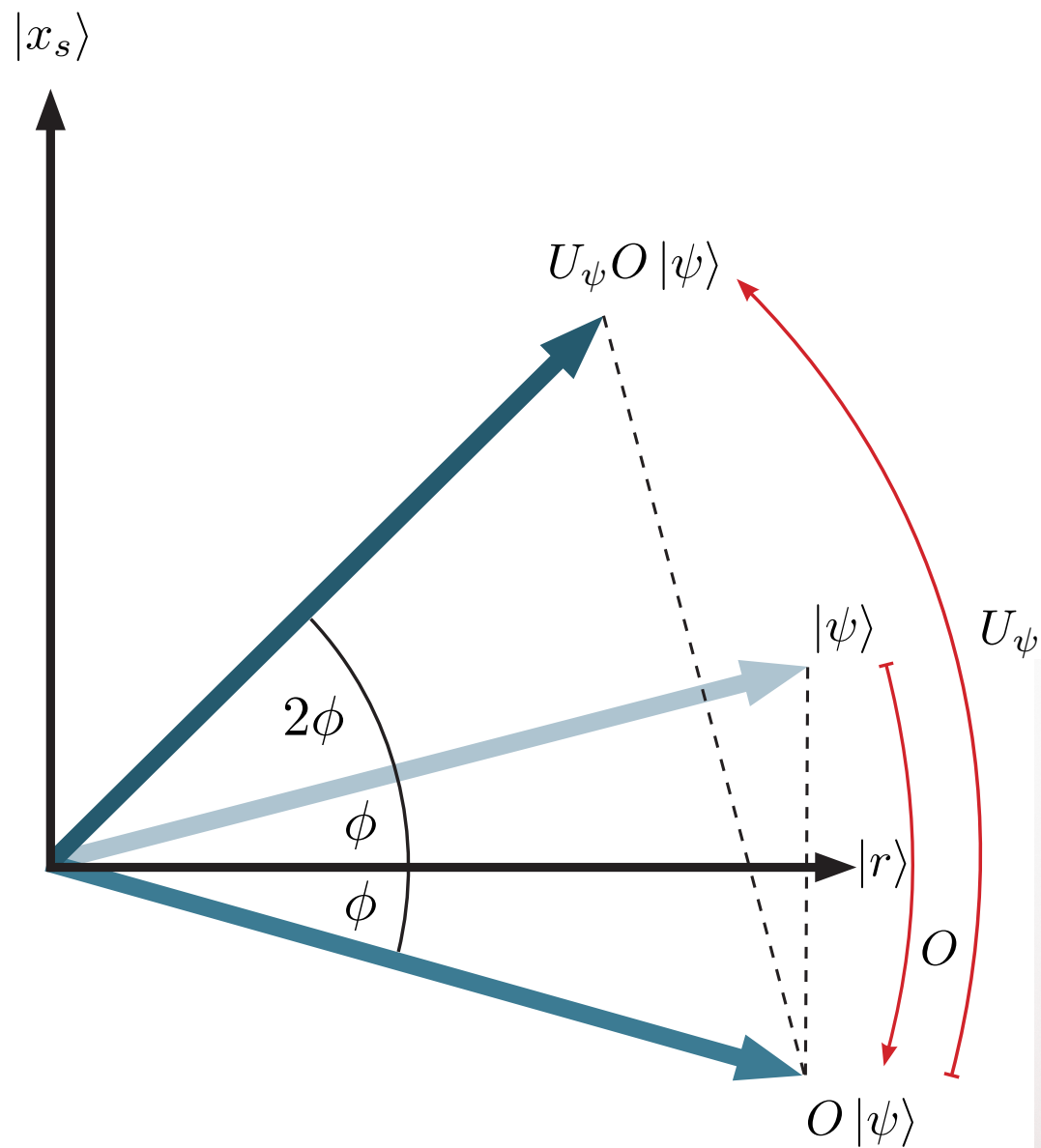
In general, $\varrho_0 = |\Psi\rangle \langle \Psi| \xrightarrow{\$} \sum_j M_j |\Psi\rangle \langle \Psi| M_j^\dagger = \varrho$

with $p(x_s) = \langle x_s | \varrho | x_s \rangle > \frac{1}{n^k}$ for $k \in \mathbb{N}, n = \log_2(N) > n_0$

Comments

- Best operation depends on witness function $f(x)$.
- *For the least structure, $f(x) = \delta_{xx_s}$, the optimal algorithm with unitary gates is Grover's search algorithm.*

Grover's Search

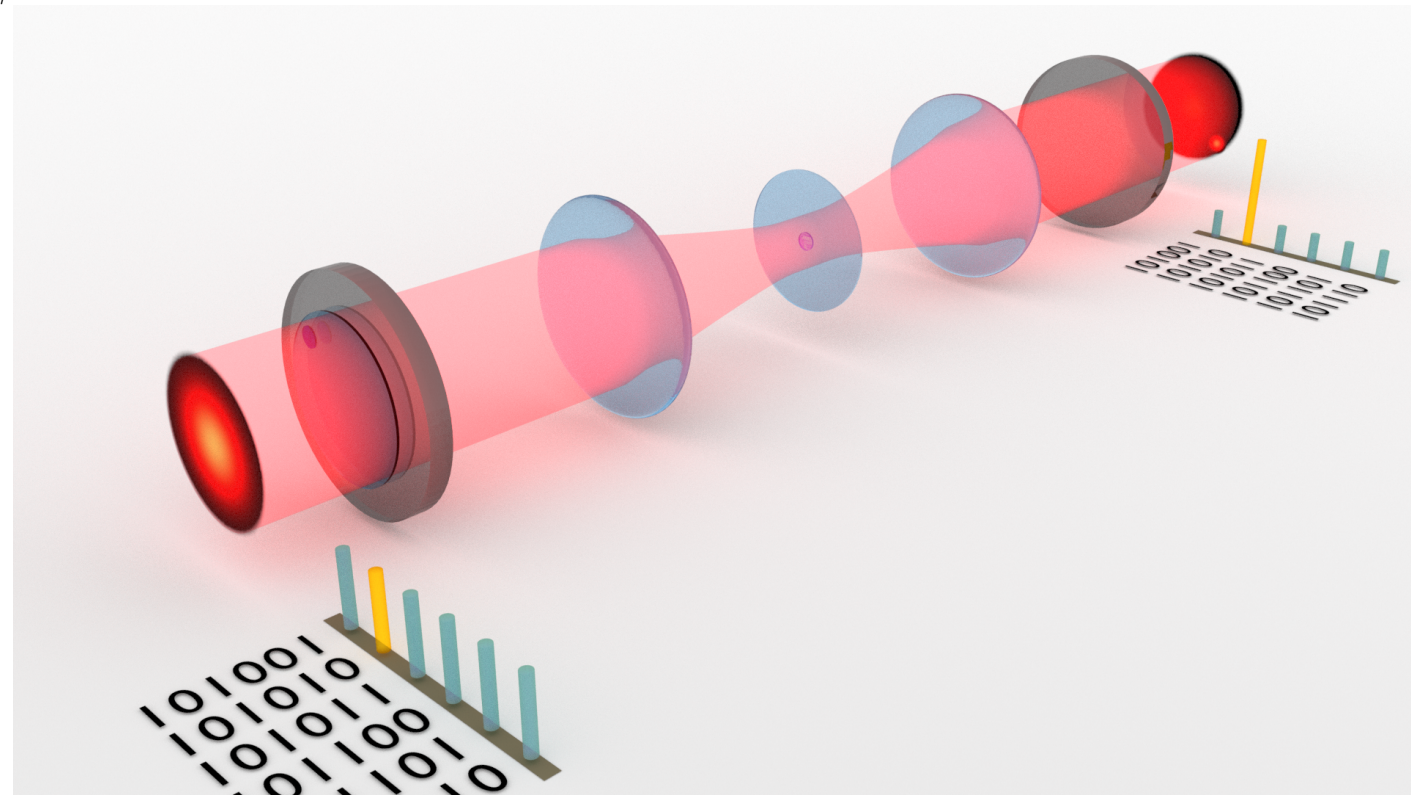


$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$$

$$|r\rangle = \frac{1}{\sqrt{N}} \sum_{x \neq x_s} |x\rangle$$

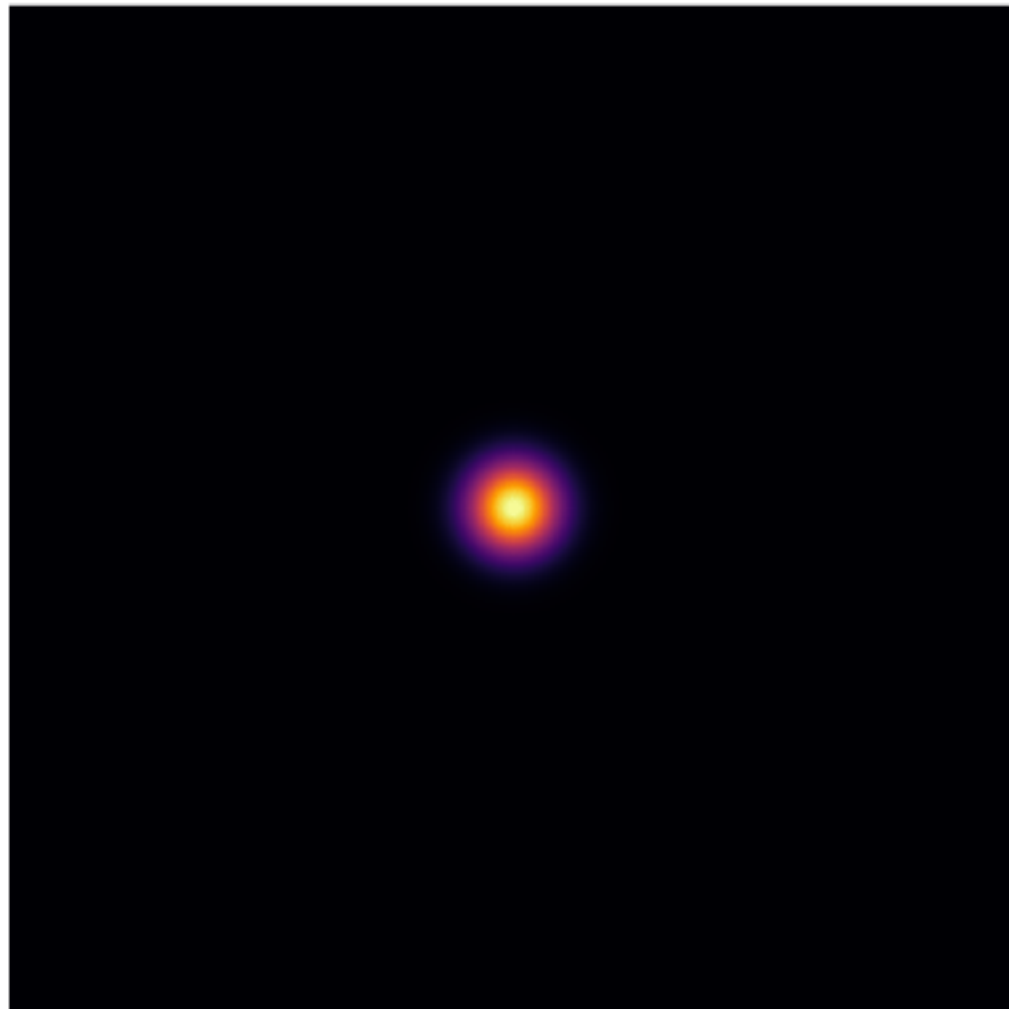
$$O = 2 |x_s\rangle \langle x_s| - \mathbb{I}$$

$$R_\psi = 2 |\psi\rangle \langle \psi| - \mathbb{I}$$

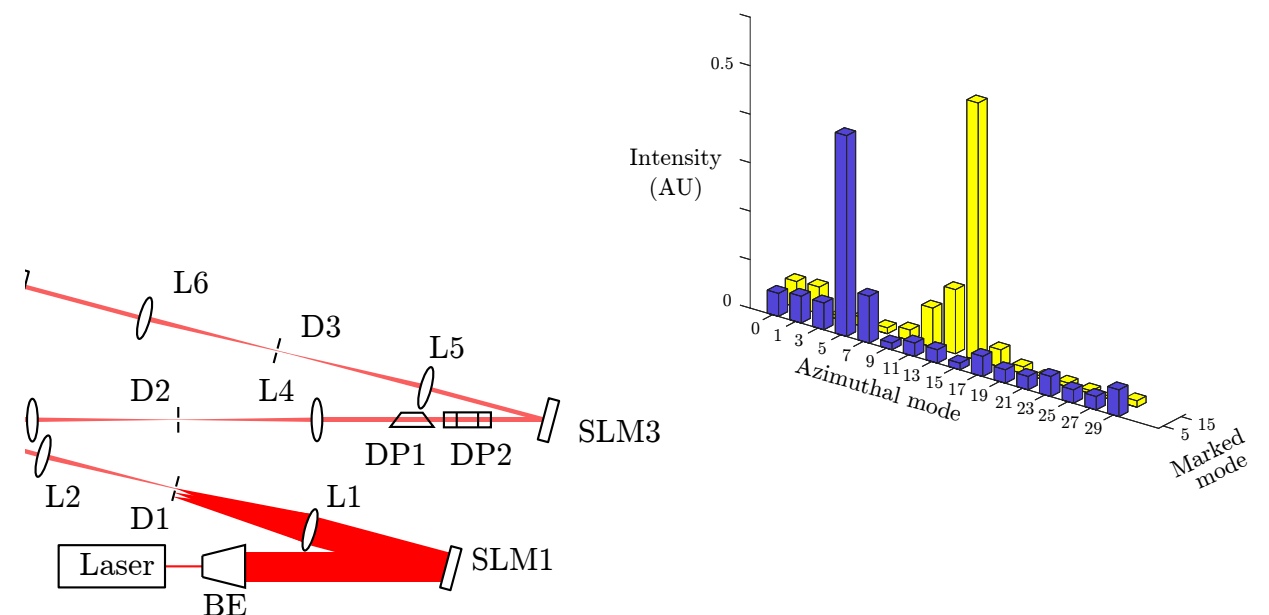
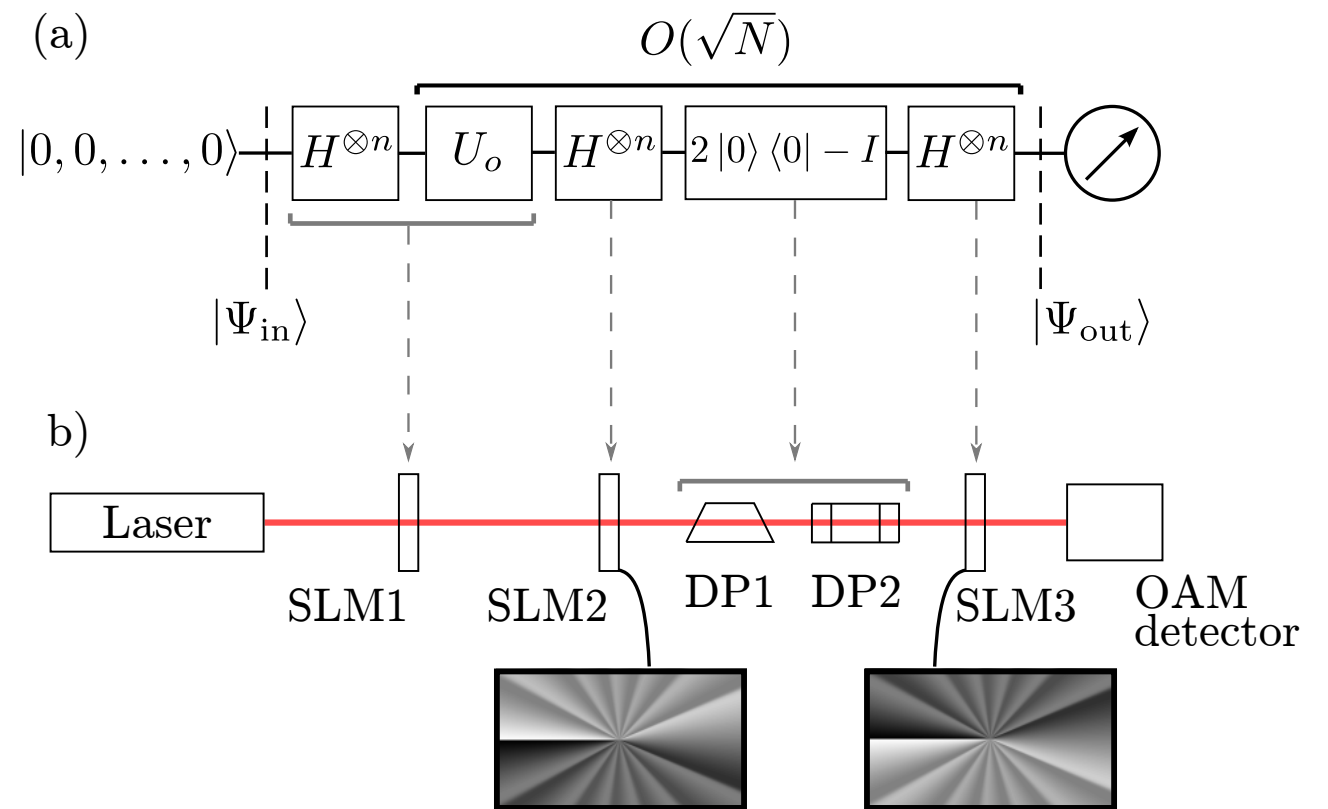


Fun with Grover in optics

Grover's Lamp
size 28000



Movie by Benjamin
Perez-Garcia



Perez-Garcia et al. Journal of Modern Optics, Volume 65, Issue 16, p.1942 (2018)